



RESEARCH METHODOLOGY

Data-Driven Property Valuation for
Metro Cebu

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QUANTITATIVE • SUPERVISED LEARNING • COMPARATIVE MODELING

APPROACH

- Supervised Learning (Regression)

DEPENDENT VARIABLE

- Property Price (₱) / Price per sqm

INDEPENDENT VARIABLES

- Structural, Locational, Macroeconomic, Text-derived features

Interpretability



Predictive Accuracy

Hedonic Regression
(Baseline)

Random Forest
(Non-Linear)

XGBoost
(High-performance)

FLOOR PRICE + CEILING PRICE + MACRO CONTEXT

Why Hybrid?

"Actual Deed of Sale data is private and inaccessible." — (Spatial Segmentation study, 2024).

Validation Layer: Human-in-the-Loop

Licensed real estate brokers (CPRE network)

- **Role 1:** Validate outliers ("Why is this Lahug property so cheap?")
- **Role 2:** Domain sanity check on key drivers vs. brokerage experience

Source	Role	Volume	Nature
BDO Foreclosures	Verified Floor Price	955 raw → ~80-100 Cebu	Conservative / Distressed
Online Listings	Market Ceiling Price	Target: 500+	Asking / Speculative
BIR Zonal Values	Administrative Benchmark	Per barangay	Static / Regulatory
BSP RPPI	Time-trend Control	Quarterly index	Macro / Cyclical

FROM RAW EXCEL TO MODEL-READY FEATURES

01

INGESTION



02

FILTERING



03

PARSING

`/ (reg) ex /`

*"Bedrooms (BR), Bathrooms (T&B),
Parking/Garage"*

04

GEOCODING



05

AUGMENTATION

Feature engineering — proximity, amenity
scores, text features

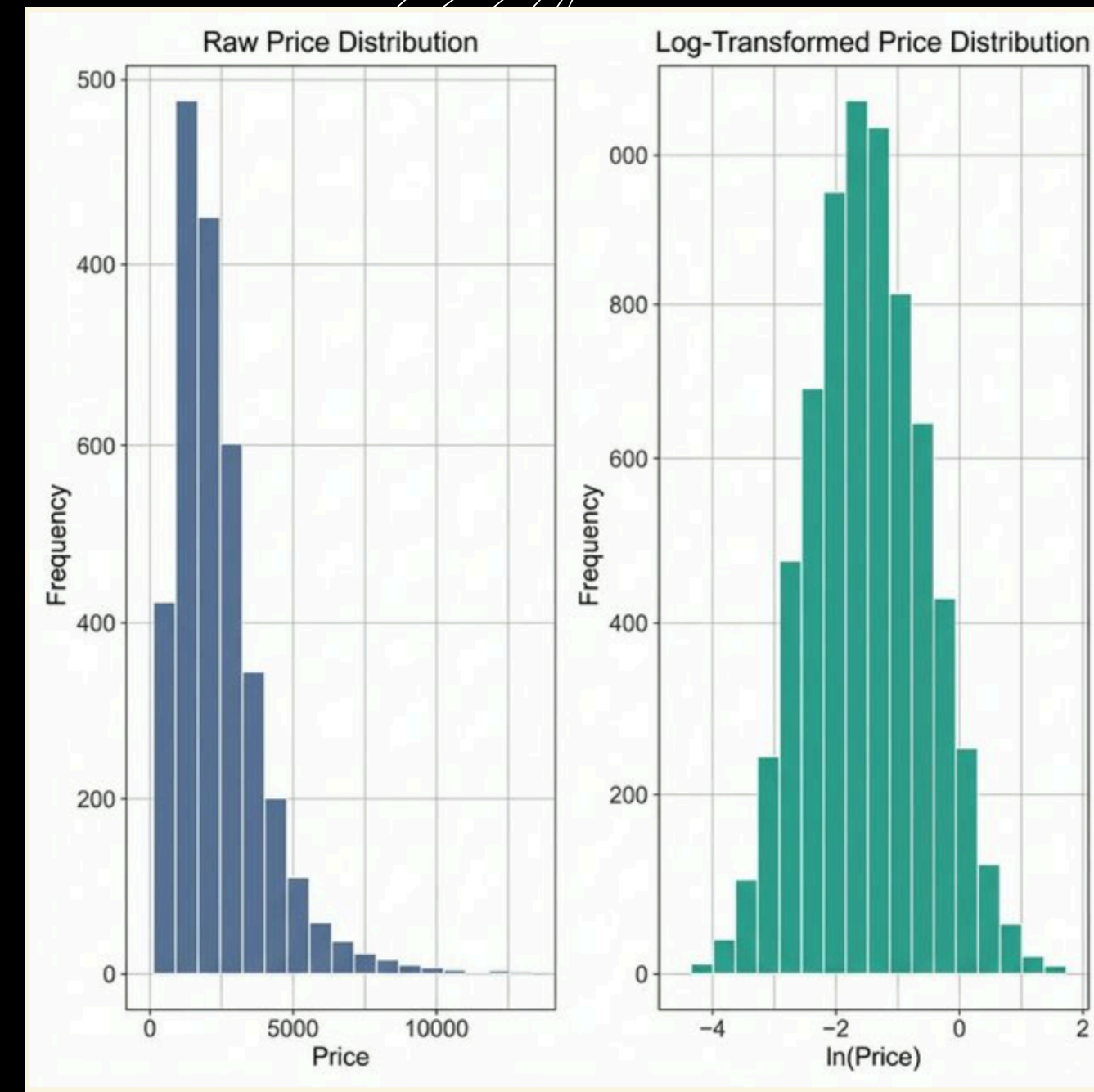
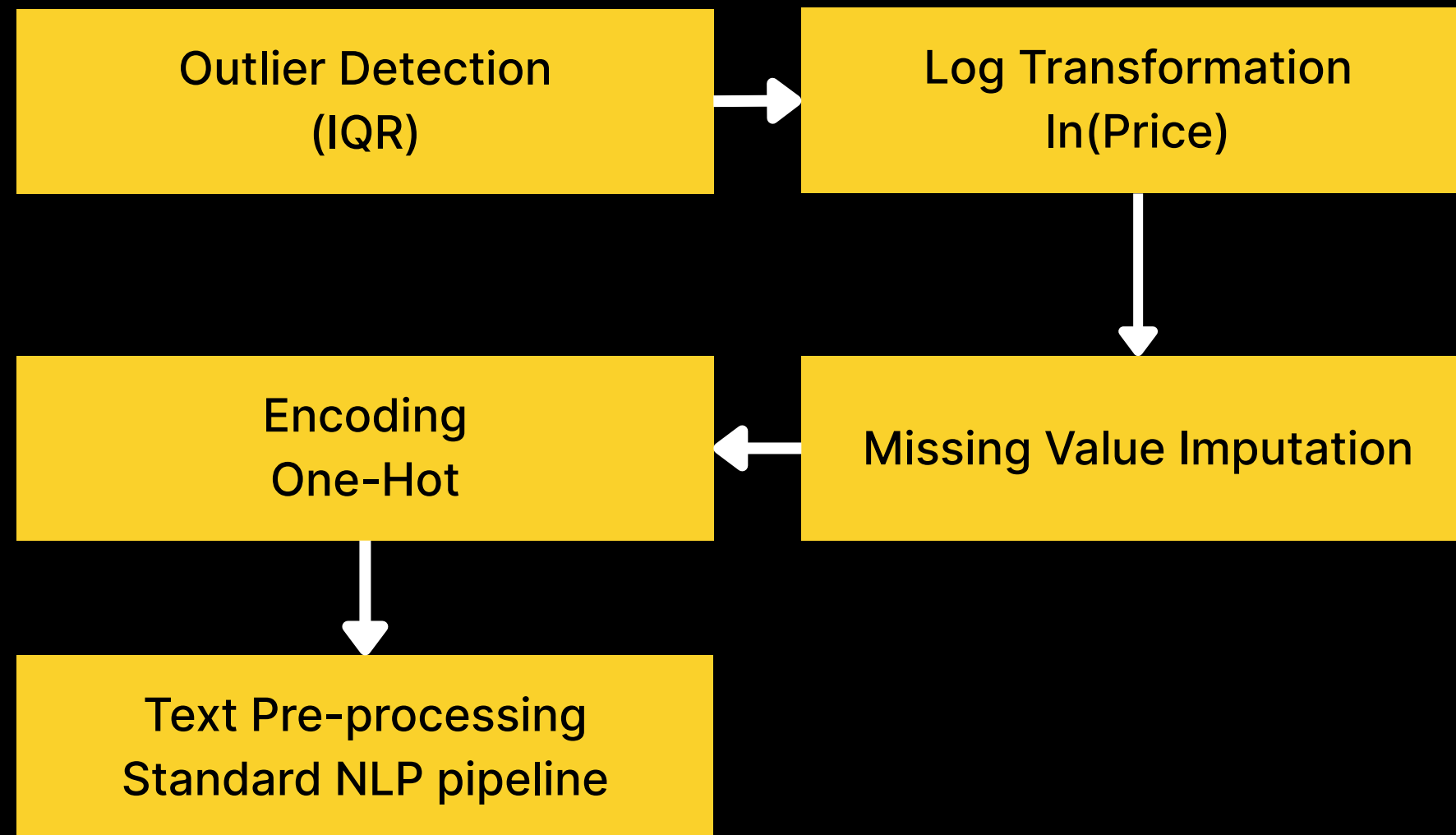


Turning raw attributes into predictive features – including text

Category	Features	Source
Structural	Lot Area, Floor Area, Bedrooms, Bathrooms, Parking, Property Type	BDO / Lamudi
Locational	Barangay, Lat/Lon coordinates	Geocoding
Proximity	Distance to Ayala, IT Park, SM Seaside, Mactan Airport	Haversine formula
<u>Text Features</u>	Keywords, TF-IDF vectors, or BERT embeddings from descriptions	NLP Pipeline
Amenity Score	Count of schools, hospitals, commercial within 1km radius	OSM / Google Places
Administrative	BIR Zonal Value per barangay	BIR schedules
Macro	BSP RPPI quarterly index (AONCR)	BSP data

CLEAN DATA IN, CLEAN PREDICTIONS OUT

PRE-PROCESSING



Price Distribution Comparison: Raw vs Log-Transformed

MODELING STRATEGY

THREE MODELS, ONE QUESTION: WHICH EXPLAINS CEBU'S PRICES BEST?

01

MULTIPLE LINEAR REGRESSION

Hedonic • Parametric

Strength: Interpretable; coefficients have economic meaning

Weakness: Assumes linearity

02

RANDOM FOREST

Ensemble (Bagging)

Strength: Handles non-linearities; robust to overfitting

Weakness: Less interpretable

03

XGBOOST

Ensemble (Boosting)

Strength: Best predictive performance on tabular data

Weakness: Hyperparameter-sensitive

EXPLORATORY EXTENSION: HYBRID LLM + ML APPROACH

Core idea: Use listing text as a feature source — not a replacement for structured data.

[Listing Text] → [Embeddings] → [XGBoost/RF] → [Price]

↑

+ structured features (sqm, location, etc.)

MAKING OUR MODEL TRANSPARENT – AND MEASURING IT AGAINST THE LITERATURE

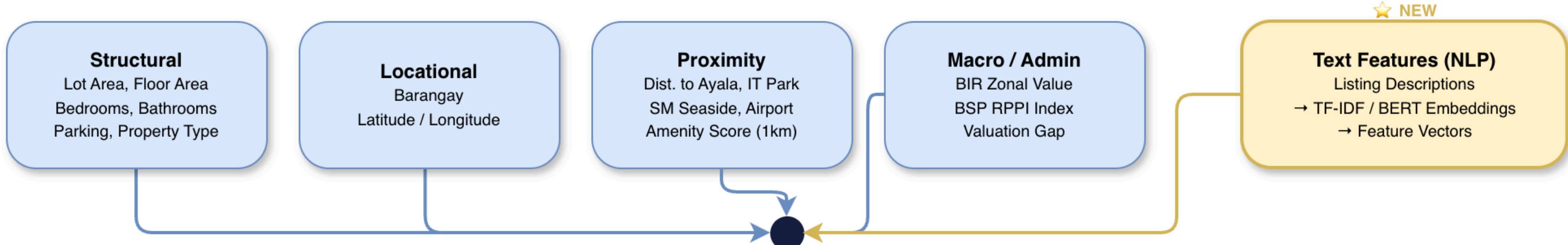
EXPLAINABILITY: SHAP VALUES

- **Why SHAP?**
 - IVS 2025 requires AVM transparency — **"no model without professional judgement can produce an IVS-compliant valuation"**
- **Global Feature Importance:** Which factors drive Metro Cebu property prices most?
- **Force Plots:** Explain individual predictions
 - "This Lahug condo is +₱1.2M due to IT Park proximity, -₱300K due to small floor area"
- **NLP Feature Attribution:** How much do text-derived features (TF-IDF/BERT) improve predictions vs. structured-only models?

Study	Context	Best MAPE	Our Target	Why This Target?
Ramolete et al. (PH, 2023)	Cavite & NCR; Lamudi; AdaBoost/XGBoost; segmented markets	10.7–21%	< 25%	They had larger, cleaner NCR data + market segmentation (87.5% benefited). Metro Cebu has fewer listings and no prior segmentation study.
Nyanda et al. (Tanzania, 2024)	Dar es Salaam; 954 obs; Boosting	0.48	Beat 48%	Comparable sample size (~955 BDO obs), but we add NLP features + online listings they didn't have.

Empirical Framework: Data-Driven Property Valuation for Metro Cebu

INDEPENDENT VARIABLES



Sources: BDO Foreclosures | Lamudi/DotProperty | BIR Zonal | BSP RPPI

